

Tab 1

Technique 1

Step 1 : Sudo ifconfig

```
RX packets 7548 bytes 5677224 (5.6 MB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 4607 bytes 1226099 (1.2 MB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

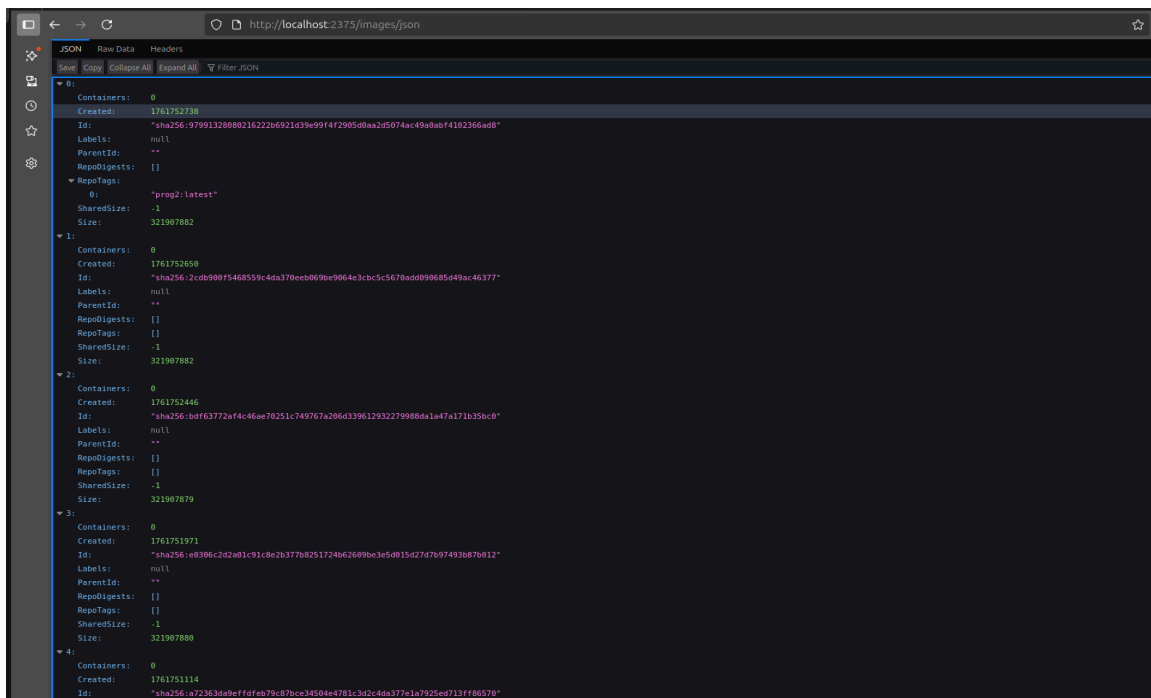
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ sudo nano /lib/systemd/system/docker.service
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ sudo systemctl daemon-reload ; systemctl restart docker
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ idconfig
Command 'idconfig' not found, did you mean:
  command 'iconfig' from deb ipmiutil (3.1.9-3)
  command 'ifconfig' from deb net-tools (2.10-0.1ubuntu4.4)
  command 'iwconfig' from deb wireless-tools (30~pre9-13.1ubuntu4)
  command 'ldconfig' from deb libc-bin (2.39-0ubuntu8.6)
Try: sudo apt install <deb name>
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.18.0.1 netmask 255.255.0.0 broadcast 172.18.255.255
    ether c2:40:cb:c3:c4:01 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 17 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1198 bytes 132970 (132.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1198 bytes 132970 (132.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlp1s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.17.6.15 netmask 255.255.240.0 broadcast 172.17.15.255
    inet6 fe80::9bc6:ad5e:d5a1:8154 prefixlen 64 scopeid 0x20<link>
    ether 14:13:33:b9:d9:8b txqueuelen 1000 (Ethernet)
    RX packets 9124 bytes 5966540 (5.9 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5155 bytes 1417012 (1.4 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$
```

Step 2 : Port Number localhost 2375



The screenshot shows a web browser window with the address bar displaying `http://localhost:2375/images/json`. The page content is a JSON array of image objects, each representing a Docker image. The JSON is displayed in a dark-themed editor with syntax highlighting. The array contains five objects, each with the following structure:

- `Containers`: 0
- `Created`: A timestamp (e.g., 1761752738).
- `Id`: A SHA256 hash (e.g., "sha256:97991328080216222b6921d39e99f4f2905d0ba2d5074ac49abaf4102366ad8").
- `Labels`: null
- `ParentId`: ""
- `RepoDigests`: []
- `RepoTags`: ["prog2:latest"]
- `SharedSize`: -1
- `Size`: 321907802

The objects are indexed 0 through 4, and each has a `Size` of approximately 321907800 bytes.

Technique 2

Step 1: Create a new folder - secondfile

```
mkdir secondfile
```

Step 2: Move into the folder

```
cd secondfile
```

Step 3: Create a Dockerfile and add the code

```
sudo nano Dockerfile
```

Step 4: Build the Dockerfile

```
sudo docker build -t secondfile .
```

Step 5: Check the Images

```
sudo docker images
```

Step 6: Run the Image - secondfile

```
sudo docker run secondfile
```

Step 7: Observe the prompt being withheld by the process & close the connection -

```
ctrl + c
```

Step 8: Re-run the Image in the detached mode

```
sudo docker run -d secondfile
```

Step 9: Observe the prompt exited

Step 10: Check the process state of docker

```
sudo docker ps
```

```

RUN apk add --update redis
CMD ["redis-server"]
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondfile$ docker build -t secondfile
ERROR: docker: 'docker buildx build' requires 1 argument

Usage: docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondfile$ docker build -t secondfile .
[+] Building 15.0s (6/6) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 97B                               0.0s
=> [internal] load metadata for docker.io/library/alpine:latest  0.2s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [1/2] FROM docker.io/library/alpine:latest@sha256:4b7ce07002c69e8f3d704a9c5d6fd3 0.0s
=> => resolve docker.io/library/alpine:latest@sha256:4b7ce07002c69e8f3d704a9c5d6fd3 0.0s
=> => sha256:4b7ce07002c69e8f3d704a9c5d6fd3053be500b7f1c69fc0d80990 9.22kB / 9.22kB 0.0s
=> => sha256:85f2b723e106c34644cd5851d7e81ee87da98ac54672b29947c052 1.02kB / 1.02kB 0.0s
=> => sha256:706db57fb2063f39f69632c5b5c9c439633fda35110e65587c5d85553f 581B / 581B 0.0s
=> [2/2] RUN apk add --update redis                             6.7s
=> => exporting to image                                          0.0s
=> => exporting layers                                           0.0s
=> => writing image sha256:5bfb94629bce15bc589e3d53b4c45e217c91cd23ec9616968ff36e77 0.0s
=> => naming to docker.io/library/secondfile                    0.0s
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondfile$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
secondfile          latest         5bfb94629bce   17 seconds ago 17.8MB
prog2               latest         979913280802   5 days ago     322MB
<none>              <none>         2cdb900f5468   5 days ago     322MB
<none>              <none>         bdf63772af4c   5 days ago     322MB
<none>              <none>         e0306c2d2a01   5 days ago     322MB
prog1               latest         a72363da9eff   5 days ago     132MB
<none>              <none>         9f1e3ce4b14c   6 days ago     445MB
<none>              <none>         1ccf0088381b   6 days ago     319MB
<none>              <none>         ae0998ac5b16   6 days ago     319MB
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondfile$ ^C
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondfile$ docker run -d secondfile
Unable to find image 'secondfile:latest' locally
docker: Error response from daemon: pull access denied for secondfile, repository does not exist or may require 'docker login': denied: requested access to the resource is denied

Run 'docker run --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondfile$ docker run -d secondfile:latest
1fc602ed8bb304b436191dbd393eab4a5d9324ebe5b7b06e29542f3ade95343d
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondfile$

```

Technique 3

Step 1: Stop the Docker service

`sudo systemctl stop docker.service`

Step 2: Move the contents of Docker to a different location

`sudo dockerd --data-root /home/deepika/mydocker`

Step 3: Start the Docker service

`sudo systemctl start docker.service`

Step 3: Start the Docker service

`sudo systemctl start docker.service`

Step 1: Check the folder creation and contents by navigating into the directory

`cd /home/deepika/mydocker`

```
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondfile$ sudo systemctl stop docker.service
Stopping 'docker.service', but its triggering units are still active:
docker.socket
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondfile$ cd ..
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ whereami
Command 'whereami' not found, but can be installed with:
sudo apt install whereami
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ whoami
mayur-shetty-c-m
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ pwd
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ sudo dockerd --data-root /home/mayur-shetty-c-m/mydocker
```

Technique - 4

Step 1: Run socat using

`sudo socat -v UNIX-LISTEN:/tmp/dockerapi.sock,fork
UNIX-CONNECT:/var/run/docker.sock &`

Step 2: To see the request and response

`sudo docker -H unix:///tmp/dockerapi.sock ps -a`

```

mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondFile$ sudo systemctl stop docker.service
Stopping 'docker.service', but its triggering units are still active:
docker.socket
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/secondFile$ cd ..
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ whereami
Command 'whereami' not found, but can be installed with:
sudo apt install whereami
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ whoami
mayur-shetty-c-m
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ pwd
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ sudo dockerd --data-root /home/mayur-shetty-c-m/.mydocker

```

Technique 5

Teq 5. Using Docker in your Browser

Teq. 1 - Open your Docker daemon to the World

```

RX packets 7548  bytes 5677224 (5.6 MB)
RX errors 0  dropped 0  overruns 0  frame 0
TX packets 4607  bytes 1226099 (1.2 MB)
TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

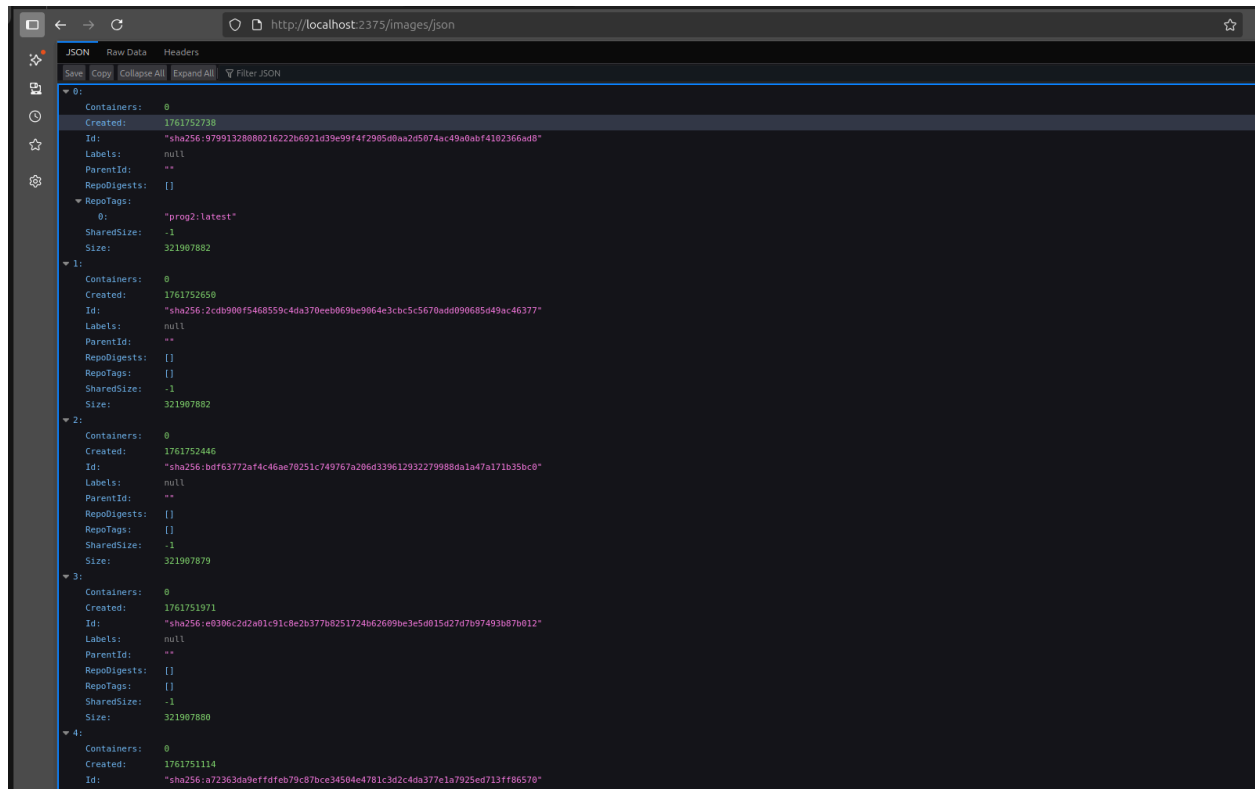
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ sudo nano /lib/systemd/system/docker.service
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ sudo systemctl daemon-reload ; systemctl restart docker
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ idconfig
Command 'idconfig' not found, did you mean:
  command 'ifconfig' from deb ipmiutil (3.1.9-3)
  command 'ifconfig' from deb net-tools (2.10-0.1ubuntu4.4)
  command 'iwconfig' from deb wireless-tools (30-pre9-13.1ubuntu4)
  command 'ldconfig' from deb libc-bin (2.39-0ubuntu8.6)
Try: sudo apt install <deb name>
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
    inet 172.18.0.1  netmask 255.255.0.0  broadcast 172.18.255.255
    ether c2:40:c0:c3:c4:01  txqueuelen 0  (Ethernet)
    RX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 0  bytes 0 (0.0 B)
    TX errors 0  dropped 17 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 1198  bytes 132970 (132.9 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 1198  bytes 132970 (132.9 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

wlp1s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 172.17.0.15  netmask 255.255.240.0  broadcast 172.17.15.255
    inet6 fe80::9bc6:ad5e:d5a1:8154  prefixlen 64  scopeid 0x20<link>
    ether 14:13:33:b9:d9:0b  txqueuelen 1000  (Ethernet)
    RX packets 9124  bytes 5966540 (5.9 MB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 5155  bytes 1417012 (1.4 MB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$

```



Technique 6

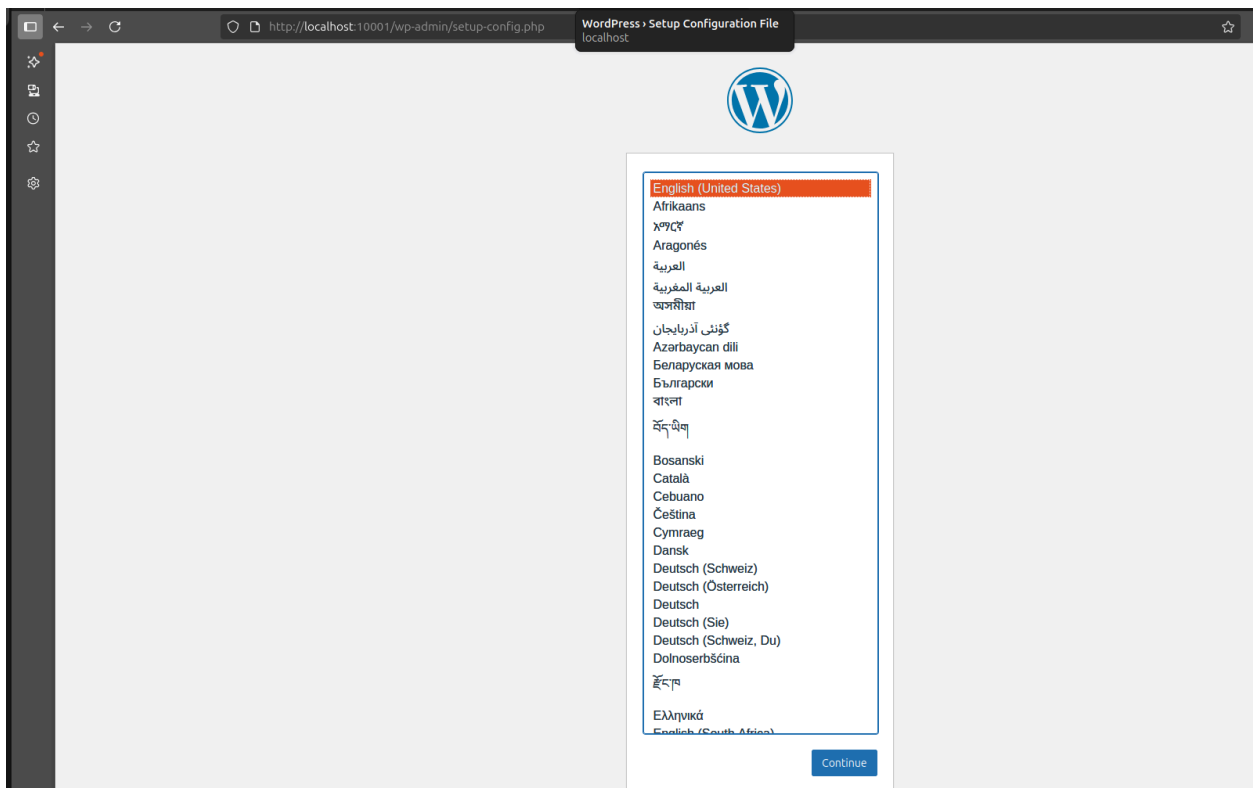
Step 1: Pull some image from Repository
`docker pull wordpress`

tep 2: Map the port 10001 to port 80
`docker run -d -p 10001:80 --name container1 wordpress`

```

mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker pull wordpress
Using default tag: latest
latest: Pulling from library/wordpress
38513bd72563: Already exists
7c587c536410: Pull complete
3262e3d480fc: Pull complete
96dfba1a7aa9: Pull complete
38a62f60c0ae: Pull complete
0c7f7dbf73bd: Pull complete
095e60baaa1c: Pull complete
5d7ef5debd8e: Pull complete
978f9bcbf3eb: Pull complete
705807c02638: Pull complete
3c631ec4c979: Pull complete
061dd099b2ba: Pull complete
cc4c6a09928c: Pull complete
75c3bd976576: Pull complete
4f4fb700ef54: Pull complete
dee1bf57e253: Pull complete
0eb92cc49677: Pull complete
74d8a97bb60c: Pull complete
5e59b25b0c46: Pull complete
bda09bfffbbd8: Pull complete
803d8813d4c6: Pull complete
d183a52dadcf: Pull complete
7e2a3357b4fb: Pull complete
28fc03424957: Pull complete
Digest: sha256:d58bf36cd0911273190beb740d8c7fad5fa30f35a4198131deb11573709ad4c1
Status: Downloaded newer image for wordpress:latest
docker.io/library/wordpress:latest
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$
docker run -d -p 10001:80 --name tech6 wordpress
1b808e0027879911899a1211fd787273d80d16e6186d9de083d66bb624fd64fc
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$

```



Technique 7

Step 1: Check the Networks - (Default: host, none and bridge)

```
docker network ls
```

Step 2: Create a user-defined network:

```
docker network create my_network
```

Step 3: Create two containers (in separate shells)

```
docker run --name container1 -it firstfile:latest
```

```
docker run --name container2 -it firstfile:latest
```

Step 3: Connect two containers

```
docker network connect my_network container1
```

```
docker network connect my_network container1
```

Step 4: Inspect the Network - my_network (find the connected containers inside the Containers)

```
docker network inspect my_network
```

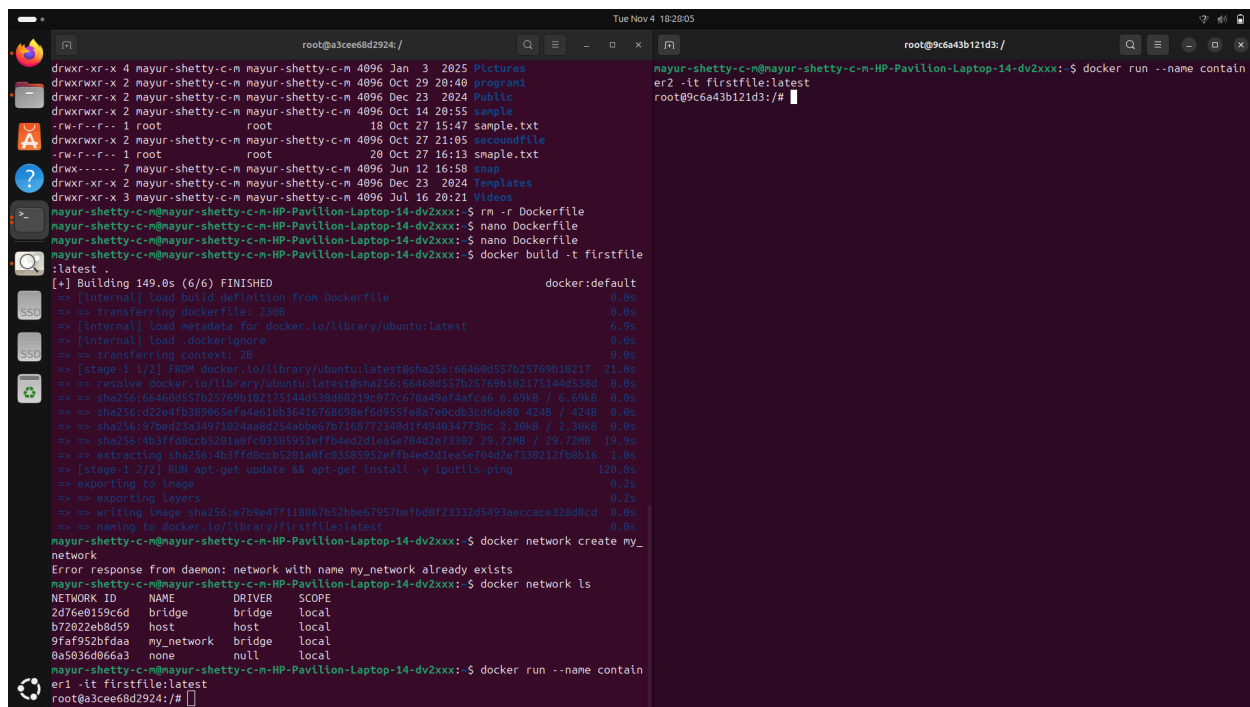
Step 4: Inspect the Network - my_network (find the connected containers inside the Containers)

```
docker network inspect my_network
```

Step 5: Install iputils-ping inside both the containers

```
apt-get install iputils-ping
```

teq 6: Ping the containers using
docker exec -it container1 ping container2



```
root@a3cee68d2924:/ # ls
drwxr-xr-x 4 mayur-shetty-c-m mayur-shetty-c-m 4096 Jan 3 2025 Pictures
drwxrwxr-x 2 mayur-shetty-c-m mayur-shetty-c-m 4096 Oct 29 20:40 program1
drwxr-xr-x 2 mayur-shetty-c-m mayur-shetty-c-m 4096 Dec 23 2024 Public
drwxrwxr-x 2 mayur-shetty-c-m mayur-shetty-c-m 4096 Oct 14 20:55 sample
-rw-r--r-- 1 root root 18 Oct 27 15:47 sample.txt
drwxrwxr-x 2 mayur-shetty-c-m mayur-shetty-c-m 4096 Oct 27 21:05 secoundfile
-rw-r--r-- 1 root root 20 Oct 27 16:13 snaple.txt
drwx----- 7 mayur-shetty-c-m mayur-shetty-c-m 4096 Jun 12 16:58 snap
drwxr-xr-x 2 mayur-shetty-c-m mayur-shetty-c-m 4096 Dec 23 2024 Templates
drwxr-xr-x 3 mayur-shetty-c-m mayur-shetty-c-m 4096 Jul 16 20:21 Videos
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ rm -r Dockerfile
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ nano Dockerfile
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ nano Dockerfile
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker build -t firstfile
:latest .
[+] Building 149.0s (6/6) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.0s
=> transferring dockerfile: 230B 0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest 0.0s
=> [internal] load .dockerignore 0.0s
=> transferring context: 2B 0.0s
=> [stage 1 1/2] FROM docker.io/library/ubuntu:latest@sha256:66460d557b25769b10217 21.0s
=> => resolve docker.io/library/ubuntu:latest@sha256:66460d557b25769b102175144d538d 0.0s
=> => sha256:66460d557b25769b102175144d538d88219c077c678a49af4afca6 6.69kB / 6.69kB 0.0s
=> => sha256:d22e4fb389065ef4a61bb36416768698efcd955fea7e0cdd3cd6de88 424B / 424B 0.0s
=> => sha256:97bed23a34971024a8dd254abbe67b7168772348d1f494034773bc 2.30kB / 2.30kB 0.0s
=> => sha256:4b37f08cc501a0fc9350592ef7b4e2d1e5e78402e71382 29.72MB / 29.72MB 19.9s
=> => extracting sha256:4b37f08cc501a0fc9350592ef7b4e2d1e5e78402e71382 1.0s
=> [stage 1 2/2] RUN apt-get update && apt-get install -y lptutils ping 120.8s
=> exporting to image 0.2s
=> exporting layers 0.2s
=> writing image sha256:a7b9e47f18867b52bbe67957befbd0f23332d5493a6ccace328d0cd 0.0s
=> naming to docker.io/library/firstfile:latest 0.0s
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker network create my_
network
Error response from daemon: network with name my_network already exists
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker network ls
NETWORK ID NAME DRIVER SCOPE
2d76e0159c6d bridge bridge local
b72022eb8d59 host host local
9faf952bfdaa my_network bridge local
0a5036d066a3 none null local
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run --name contain
er1 -it firstfile:latest
root@a3cee68d2924:/#
```

Technique 8

Step 1: Check the Networks - (Default: host, none and bridge)

docker network ls

Step 2: Pull images of mysql and apache

docker pull mysql/mysql-server:5.6

Step 3: Run mysql and apache specifying a port number and

docker run -p 3306:3306 --name mysql -e

MYSQL_ROOT_PASSWORD=root123 -d

mysql/mysql-server:5.6

docker run -tid -p 80:80 --name apache2 --link mysql

nimmis/apache-php5

Step 4: Check the docker process state

`docker ps`

Step 5: Check the networks of both containers

`docker inspect mysql`

Step 6: Ping mysql to verify then

Ping mysql

```
4f4fb700ef54: Pull complete
b4fc1b614181: Pull complete
fa2c8cfe7774: Pull complete
61453b567d7a: Pull complete
Digest: sha256:15c7bc671aa8c6b7ac23015bbf91efd93404c331f7e27e685fdeb6a0b4fbcd2a
Status: Downloaded newer image for nimmis/apache-php5:latest
docker.io/nimmis/apache-php5:latest
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run -p 3306:3306 -
-name mysql -e MYSQL_ROOT_PASSWORD=root123 -d mysql/mysql-server:5.6
f97d30fde170910e5a326673412d682d3028166cc5f1459f7c24d946849d045b
docker: Error response from daemon: failed to set up container networking: driver failed pr
ogramming external connectivity on endpoint mysql (8de16df68130af43f46f96fa2c2e310627dcc6a6
de1bce742ff0ba6ebdc8cd53): failed to bind host port for 0.0.0.0:3306:172.18.0.2:3306/tcp: a
ddress already in use

Run 'docker run --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run -tid -p 80:80
--name apache2 --link mysql nimmis/apache-php5
9cd1d8f4766a579ede7bd933353cd328db5950a39dc7e873a1ae2fb1b32259df
docker: Error response from daemon: Cannot link to a non running container: /mysql AS /apac
he2/mysql

Run 'docker run --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$
```

```
9cd1d8f4766a579ede7bd933333cd328db5950a39dc7e873a1ae2fb1b32259d1
docker: Error response from daemon: Cannot link to a non running container: /mysql AS /apac
he2/mysql
```

Run 'docker run --help' for more information

```
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker ps
docker: command not found
```

```
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
--------------	-------	---------	---------	--------	-------	-------

```
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run -p 3307:3306 -
-name mysql -e MYSQL_ROOT_PASSWORD=root123 -d mysql/mysql-server:5.6
```

```
docker: Error response from daemon: Conflict. The container name "/mysql" is already in use
by container "f97d30fde170910e5a326673412d682d3028166cc5f1459f7c24d946849d045b". You have
to remove (or rename) that container to be able to reuse that name.
```

Run 'docker run --help' for more information

```
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
--------------	-------	---------	---------	--------	-------	-------

```
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
--------------	-------	---------	---------	--------

S	PORTS	NAMES
---	-------	-------

9cd1d8f4766a	nimmis/apache-php5	"/my_init"	About a minute ago	Created
--------------	--------------------	------------	--------------------	---------

f97d30fde170	mysql/mysql-server:5.6	"/entrypoint.sh mysql..."	About a minute ago	Created
--------------	------------------------	---------------------------	--------------------	---------

9c6a43b121d3	firstfile:latest	"bash"	29 minutes ago	Exited
--------------	------------------	--------	----------------	--------

a3cee68d2924	firstfile:latest	"bash"	29 minutes ago	Exited
--------------	------------------	--------	----------------	--------

1b808e002787	wordpress	"docker-entrypoint.s..."	2 hours ago	Exited
--------------	-----------	--------------------------	-------------	--------

```
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run 9cd1
```

```
Unable to find image '9cd1:latest' locally
```

```
docker: Error response from daemon: pull access denied for 9cd1, repository does not exist
or may require 'docker login': denied: requested access to the resource is denied
```

Run 'docker run --help' for more information

```
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$
```

766a

Error response from daemon: Cannot link to a non running container: /mysql AS /apache2/mysql

mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~\$ docker run -tid -p 8080:80 --name apache2 --link mysql:mysql nimms/apache-php5

Unable to find image 'nimms/apache-php5:latest' locally

docker: Error response from daemon: pull access denied for nimms/apache-php5, repository does not exist or may require 'docker login': denied: requested access to the resource is denied

Run 'docker run --help' for more information

mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~\$ docker rm -f mysql
mysql

mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~\$ docker run -p 3307:3306 --name mysql -e MYSQL_ROOT_PASSWORD=root123 -d mysql/mysql-server:5.6
28fbb69b965ebb475a54ad694e930957bac5db6b7030cd69bf7d41f936a2e5e2

mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~\$ docker run -tid -p 8080:80 --name apache2 --link mysql:mysql nimms/apache-php5

Unable to find image 'nimms/apache-php5:latest' locally

docker: Error response from daemon: pull access denied for nimms/apache-php5, repository does not exist or may require 'docker login': denied: requested access to the resource is denied

Run 'docker run --help' for more information

mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~\$ docker ps -a

CONTAINER ID	IMAGE	PORTS	COMMAND	CREATED	STATUS
28fbb69b965e	mysql/mysql-server:5.6	0.0.0.0:3307->3306/tcp	"/entrypoint.sh mysq..."	16 seconds ago	Up 15 seconds (health: starting)
9cd1d8f4766a	nimmis/apache-php5		"/my_init"	3 minutes ago	Created
9c6a43b121d3	firstfile:latest		"bash"	31 minutes ago	Exited (1)
a3cee68d2924	firstfile:latest		"bash"	32 minutes ago	Exited (1)
1b808e002787	wordpress		"docker-entrypoint.s..."	3 hours ago	Exited (0)

mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~\$

```
server:5.6
docker pull php:7.4-apache
5.6: Pulling from mysql/mysql-server
Digest: sha256:4b2262a86a7fa9010ffa8153eb0fe751feba83a0dac9aa594c940726ca01cb99
Status: Image is up to date for mysql/mysql-server:5.6
docker.io/mysql/mysql-server:5.6
7.4-apache: Pulling from library/php
a603fa5e3b41: Pull complete
c428f1a49423: Pull complete
156740b07ef8: Pull complete
fb5a4c8af82f: Pull complete
25f85b498fd5: Pull complete
9b233e420ac7: Pull complete
fe42347c4ecf: Pull complete
d14eb2ed1e17: Pull complete
66d98f73acb6: Pull complete
d2c43c5efbc8: Pull complete
ab590b48ea47: Pull complete
80692ae2d067: Pull complete
05e465aaa99a: Pull complete
Digest: sha256:c9d7e608f73832673479770d66aacc8100011ec751d1905ff63fae3fe2e0ca6d
Status: Downloaded newer image for php:7.4-apache
docker.io/library/php:7.4-apache
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run -p 3306:3306 -
-name mysql -e MYSQL_ROOT_PASSWORD=root123 -d mysql/mysql-server:5.6
docker: Error response from daemon: Conflict. The container name "/mysql" is already in use
by container "28fbb69b965ebb475a54ad694e930957bac5db6b7030cd69bf7d41f936a2e5e2". You have
to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$
```

```

ogramming external connectivity on endpoint mysql (2d99904b1888a0d9817ab189d840a1a7bca4233a
af6d647a6ff28808dd2ce187): failed to bind host port for 0.0.0.0:3306:172.18.0.2:3306/tcp: a
ddress already in use

Run 'docker run --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker ps -a
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS
PORTS              NAMES
46997b146901        mysql/mysql-server:5.6   "/entrypoint.sh mysq..."   17 seconds ago     Created
9cd1d8f4766a        nimmis/apache-php5      "/my_init"           9 minutes ago      Created
9c6a43b121d3        firstfile:latest       "bash"               37 minutes ago     Exited (1
29) 35 minutes ago   container2
a3cee68d2924        firstfile:latest       "bash"               38 minutes ago     Exited (1
29) 35 minutes ago   container1
1b808e002787        wordpress             "docker-entrypoint.s..."   3 hours ago        Exited (0
) 2 hours ago       tech6
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ sudo systemctl stop mysql

[sudo] password for mayur-shetty-c-m:
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run -p 3306:3306 -
-name mysql -e MYSQL_ROOT_PASSWORD=root123 -d mysql/mysql-server:5.6
docker: Error response from daemon: Conflict. The container name "/mysql" is already in use
by container "46997b146901cd94fb1d1f405147504d12445dc24d70136551427b0953803ca7". You have
to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run -p 3307:3306 -
-name mysql -e MYSQL_ROOT_PASSWORD=root123 -d mysql/mysql-server:5.6
docker: Error response from daemon: Conflict. The container name "/mysql" is already in use
by container "46997b146901cd94fb1d1f405147504d12445dc24d70136551427b0953803ca7". You have
to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ Status: Up X seconds
Status:: command not found
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$

```

Technique 9

Step 1: To create and make the registry available

```
docker run -d -p 5000:5000 -v $HOME/registry:/var/lib/registry
registry:2
```



```

to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run -p 3307:3306 -
-name mysql -e MYSQL_ROOT_PASSWORD=root123 -d mysql/mysql-server:5.6
docker: Error response from daemon: Conflict. The container name "/mysql" is already in use
by container "46997b146901cd94fb1d1f405147504d12445dc24d70136551427b0953803ca7". You have
to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ Status: Up X seconds
Status:: command not found
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker stop mysql
docker rm mysql
mysql
mysql
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS         NAMES
9cd1d8f4766a   nimmis/apache-php5                 "/my_init"              11 minutes ago Created
apache2
9c6a43b121d3   firstfile:latest                  "bash"                  39 minutes ago Exited (129)
37 minutes ago container2
a3cee68d2924   firstfile:latest                  "bash"                  40 minutes ago Exited (129)
37 minutes ago container1
1b808e002787   wordpress                         "docker-entrypoint.s..." 3 hours ago    Exited (0) 2
hours ago    tech6
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run -p 3307:3306 -
-name mysql -e MYSQL_ROOT_PASSWORD=root123 -d mysql/mysql-server:5.6
f8192874bd637e6d856dd6046c8fdd66d3a9d3f6be7491904a43fc5846368d
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$ docker run -tid -p 8080:8
0 --name apache2 --link mysql:mysql php:7.4-apache
docker: Error response from daemon: Conflict. The container name "/apache2" is already in u
se by container "9cd1d8f4766a579ede7bd933353cd328db5950a39dc7e873a1ae2fb1b32259df". You hav
e to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~$

```

Technique 10

Step 1: Use the terminal to search the Docker Image

docker search node

Step 2: Run the Docker Image

docker pull node


```
Tue Nov 4 19:12:48
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavillon-Laptop-14-dv2xxx:~$ docker run -d -p 5000:5000 -v $HOME/.registry:/var/lib/registry registry:2
Unable to find image 'registry:2' locally
2: Pulling from library/registry
44cf07d57ee4: Pull complete
bbbdd6c6894b: Pull complete
8e82f80af0de: Pull complete
3493bf46cdec: Pull complete
6d464ea18732: Pull complete
Digest: sha256:a3d8aaa63ed9681a684f1dea0aa03f100d5895b6a58ace528858a7b332415373
Status: Downloaded newer image for registry:2
f97e7764d42056b04177201fb2975414dc740990c9a4f55574159fb1055f792f
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavillon-Laptop-14-dv2xxx:~$
```

```
Tue Nov 4 19:19:36
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavillon-Laptop-14-dv2xxx:~$ docker search node
NAME                                DESCRIPTION                                     STARS     OFFICIAL   [OK]
node                                Node.js is a JavaScript-based platform for s... 14042
bitnami/node                        Bitnami Secure Image for node                  83
cing/node                           The CircleCI Node.js Docker Convenience Imag... 26
circleci/node                       Node.js is a JavaScript-based platform for s... 135
kindest/node                        https://sigs.k8s.io/kind node image           116
okteto/node                         2
chainguard/node                     Build, ship and run secure software with Cha... 1
sitespeedio/node                   Node base template                             3
eclipse/node                       Node 0.12.9                                    1
corpusops/node                    https://github.com/corpusops/docker-images/    0
rootpublic/node                   0
setuptools/node                    Docker images to run setup-php GitHub Action   0
ubuntu/node                        Ubuntu-based Node.js image for server-side a... 4
activestate/node                   ActiveState's customizable, low-to-no vulner... 12
joxit/node                         Slim node docker with some utils for dev       1
alpine/node                        5
treehouses/node                    2
vmware/node                        Node.js base built on top of Photon OS         0
wayofdev/node                     0
vulhub/node                        0
systemsdk/node                     Docker environment with node 16 for Laravel/... 0
openizr/node                       Safer, non-root, nodeJS environment            0
openeuler/node                    0
presearch/node                     Run a search node in Presearch's decentraliz... 26
iron/node                          Tiny Node image                                28
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavillon-Laptop-14-dv2xxx:~$ docker pull node
Using default tag: latest
latest: Pulling from library/node
cd01949e3ebd: Pull complete
f279d33abc52: Pull complete
5daa5a418829: Pull complete
da3556d62e9b: Pull complete
2364b52a1d84: Pull complete
0da8ce6a0fb1: Pull complete
33a268bbaa65: Pull complete
133395198c18: Pull complete
Digest: sha256:e524a871ad29ac5fa38b840667a6590f42bc14eb052047815faa8ba421c52b93
Status: Downloaded newer image for node:latest
docker: to/library/node:latest
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavillon-Laptop-14-dv2xxx:~$
```

Tab 2

